

Section Id :	64065338348
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	19
Number of Questions to be attempted :	19
Section Marks :	45
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065380570
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 1 Question Id : 640653564614 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531887321. YES

6406531887322. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065380571
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 2 Question Id : 640653564615 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What brought the first industrial revolution?

Options :

6406531887323. ✔ Mechanization

6406531887324. ✖ Electric Power

6406531887325. ✖ Computer

6406531887326. ✖ Robotics

Question Number : 3 Question Id : 640653564617 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

The Production rate of an assembly line is determined by its

Options :

6406531887331. ✔ Slowest station

6406531887332. ✖ Fastest station

6406531887333. ✖ Idle station

6406531887334. ✖ None of these

Question Number : 4 Question Id : 640653564623 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is not a part of the 7 principles of IoT:

Options :

6406531887345. ✖ Big Analog Data

6406531887346. ✔ Quality Control

6406531887347. ✖ Constant connection

6406531887348. ✖ Real time

6406531887349. ✖ Spectrum of insights

Question Number : 5 Question Id : 640653564637 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is the decision variable X_{ij} denotes in a Traveling Salesman Problem (TSP)

Options :

6406531887387. ✖ If the sales visits node-j from node-i (binary)

6406531887388. ✔ If the sales visits node-j immediately after visiting node-i (binary)

6406531887389. ✖ The quantity of an item transferred between node-i and node-j (positive,integer)

Sub-Section Number : 3

Sub-Section Id : 64065380572

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 6 Question Id : 640653564616 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is an advantage of Process Layout?

Options :

- 6406531887327. ✖ Low material handling
- 6406531887328. ✖ Total production time per unit is small
- 6406531887329. ✖ Low training cost for the operators
- 6406531887330. ✔ It can accommodate high variety of products

Question Number : 7 Question Id : 640653564618 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

What are the inputs for an order-up-to-level calculation?

Options :

- 6406531887335. ✔ Sales forecast
- 6406531887336. ✔ Lead time
- 6406531887337. ✔ On-hand inventory
- 6406531887338. ✔ Safety stock

Question Number : 8 Question Id : 640653564626 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

What is Digitalization? Choose all that are applicable

Options :

6406531887358. ✖ Pure Analog to Digital Conversion

6406531887359. ✔ Use of digital technologies to change the processes

6406531887360. ✔ Process of moving towards a digital business

6406531887361. ✖ Mere transformation – encoded in digital format

Question Number : 9 Question Id : 640653564635 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true? Choose all that are applicable

Options :

6406531887380. ✔ Euclidean distance measures distance as the straight line between 2 points.

6406531887381. ✔ Metropolitan distance measures distance with respect to the road availability

6406531887382. ✖ None of these

Question Number : 10 Question Id : 640653564636 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Select the correct aims of a routing problem

Options :

- 6406531887383. ✓ To visit all locations
- 6406531887384. ✓ To attain minimal travel distance
- 6406531887385. ✓ To attain minimal use of resources
- 6406531887386. ✓ Demand at all locations is satisfied.

Sub-Section Number :	4
Sub-Section Id :	64065380573
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653564619 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Question Numbers : (11 to 12)

Question Label : Comprehension

The ten activities specified below are to be performed to make a product. Then answer the given subquestions.

Activity	Activity Time (in min)	Preceding Activity
A1	9	
A2	5	
A3	18	
A4	10	
A5	6	
A6	8	A1, A3
A7	16	A1, A4
A8	19	A3, A5
A9	5	A2
A10	13	A8, A9

Table Indicating Activity Information

Sub questions

Question Number : 11 Question Id : 640653564620 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are an allowable grouping of activities to produce the product if the target is to produce a product every 20 minutes?

Options :

6406531887339. ✖ (A3)→(A1,A4,A2)→ (A8)→(A10,A9)→(A7)→Output

6406531887340. ✔ (A3)→(A1,A4)→(A2,A5,A6)→(A8)→(A10,A9)→(A7)→Output

6406531887341. ✖ (A3,A2)→(A1,A4)→(A5,A6)→(A8)→(A10,A9)→(A7)→Output

6406531887342. ✖ (A2,A5,A6) → (A1,A4)→(A3)→(A8)→(A10,A9)→(A7)→Output

Question Number : 12 Question Id : 640653564621 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 2

Question Label : Short Answer Question

What is the balance delay for the line (or lines) identified in the main question?

(Enter your answer in % rounded to two decimal places, without the percentage symbol. Eg. If you compute the answer as 0.10005, then enter 10.01 as your answer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.10 to 9.20

Sub-Section Number :	5
Sub-Section Id :	64065380574
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 13 Question Id : 640653564622 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 2

Question Label : Short Answer Question

Currently, a company is using a layout which consists of 9 workstations. 200 sandwiches are produced in a day on this layout. A typical day consists of a single shift which runs for 10 hours. The sandwich-making process consists of 9 activities (the information is provided in the table below).

Activity ID	Activity Description	Activity Time (in seconds)	Preceding Activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Cutting	20	C
F	Butter applying	5	E
G	Vegetable filling	5	F, E
H	Sandwich baking	15	G
I	Sandwich packing	5	H

What is the current efficiency of the line where each activity is performed on a separate workstation (enter only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

11.10 to 11.12

Sub-Section Number : 6

Sub-Section Id : 64065380575

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 14 Question Id : 640653564624 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is Digitization?

Options :

6406531887350.

✔ Conversion of physical 2D drawings to CAD drawings

6406531887351. ✖ Creating PLC logic in a microprocessor-based system to process data and identify insights

6406531887352. ✖ Providing real-time maintenance schedule based on real-time machine performance

6406531887353. ✖ None of these

Sub-Section Number : 7
Sub-Section Id : 64065380576
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 15 Question Id : 640653564625 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Sky Airlines has to decide its plan for allowing booking a flight between Chennai and Bengaluru. The aircraft has a capacity of 100 (identical seats). The price for economy passengers is INR 1000 per seat whereas for business passengers it is INR 5000 per seat. If the demand for business passengers is uniformly distributed between [10,40], what is the protection level?

Options :

6406531887354. ✔ 34

6406531887355. ✖ 32

6406531887356. ✖ 68

6406531887357. ✖ 70

Sub-Section Number : 8
Sub-Section Id : 64065380577
Question Shuffling Allowed : Yes

Is Section Default? :	null
Question Number : 16 Question Id : 640653564627 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0	
Correct Marks : 6	
Question Label : Multiple Choice Question	
Match the following columns by selecting the correct name from Column B for the labels in column A which are depicted in the ' Figure: EOQ model ' below:	
Options :	
6406531887362. ✖ A:Cost; B:Quantity; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)	
6406531887363. ✔ A:Cost; B:Quantity; C:Total cost; D:Order cost; E:Carrying cost; F:EOQ(Q*)	
6406531887364. ✖ A:Quantity; B:Cost; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)	
6406531887365. ✖ A:Quantity; B:Cost; C:Total Cost; D:Order cost; E:Carrying cost; F:EOQ(Q*)	
6406531887366. ✖ A:Cost; B:EOQ(Q*); C:Total Cost; D:Order cost; E:Carrying cost; F:Quantity	
6406531887367. ✖ A:Total cost; B:EOQ(Q*); C:Cost; D:Order cost; E:Carrying cost; F:Quantity	
6406531887368. ✖ A:Total cost; B:Quantity; C:Cost; D:Carrying cost; E:Order cost; F: EOQ(Q*)	
Sub-Section Number :	9
Sub-Section Id :	64065380578
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 17 Question Id : 640653564628 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

At EOQ, the carrying cost is

Options :

6406531887369. ✖ Minimum

6406531887370. ✔ Equal to the order cost

6406531887371. ✖ Lower than the order cost

6406531887372. ✖ Higher than the order cost

Question Number : 18 Question Id : 640653564634 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets on the neighbouring shelves. He picked a few milk-powder packets. However, these were not part of his list. He was searching for Good-day biscuits. He could identify only one packet on the shelf. However, there were a lot of Sun-feast biscuits lying on the next shelf. So, he picked ten Sun-feast packets. Now map the scenarios with the items below:

Product	Scenario
P1) Tea : Coffee	S1) Complementary
P2) Tea: milk-powder packet	S2) Substitution
P3) Good-day biscuits: Sun-feast biscuits	S3) Inventory dependant demand
	S4) Impulse buying

Options :

6406531887377. ✖ P1-S1, P2-S3, P3-S4

6406531887378. ✔ P1-S2, P2-S1, P3-S3

6406531887379.

✖ P1-S2, P2-S1, P3-S4

Sub-Section Number : 10
Sub-Section Id : 64065380579
Question Shuffling Allowed : No
Is Section Default? : null

Question Id : 640653564629 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (19 to 22)

Question Label : Comprehension

Limat Ltd. wants to determine its order quantity for one of its components. The annual consumption for the component is 80,000 units and its purchasing price is Rs. 50 per unit. The cost of placing an order is Rs. 1,200 and carrying cost per unit is 6% of the purchasing price.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 19 Question Id : 640653564630 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the Economic Order Quantity (EOQ).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8000

Question Number : 20 Question Id : 640653564631 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the number of orders per year.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 21 Question Id : 640653564632 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the annual order cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12000

Question Number : 22 Question Id : 640653564633 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the annual inventory carrying cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12000

Sub-Section Number :	11
Sub-Section Id :	64065380580
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 23 Question Id : 640653564638 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?

Options :

- 6406531887390. ✖ Subtour elimination
- 6406531887391. ✔ Any capacity restrictions
- 6406531887392. ✔ A specific starting point
- 6406531887393. ✔ Multiple salesman